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Post-quantum cryptography for Multiple high-resolution millimeter wave images for enhanced security in IOT Networks

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***Abstract*—In the current era, the exponential rise in security threats is a significant concern, extending beyond the realm of the Internet of Things (IoT). The data being transferred and stored requires a robust encryption and decryption mechanism. The existing system utilizes the SHA algorithm and various encryption methods to secure image data. However, with the evolution of next-generation networks, quantum techniques pose a potential threat to decrypting existing cryptographic methodologies. These quantum techniques can extract data from the internet today and process it in the future. To address this looming cybersecurity challenge, this paper proposes a novel technique—a post-quantum cryptographic method designed to withstand quantum attacks in the near future. Specifically, the paper introduces a wavelength multiplexing embedded post-quantum technique capable of encrypting multiple high-resolution Millimeter-wave images. Each image, representing a pixel value of 4160 horizontally and 3120 vertically, contributes to the overall security of the method. The proposed wavelength multiplexing method is compared with the BBM92 protocol in terms of processing speed, confidentiality, integrity, and non-repudiation. Post quantum cryptographic method works in the quantum computers are fast, reliable in qubits ensures authentication, verify the identity of a system in IOT networks make the scope to support the future networks working in quantum environment and simulation environment shows a promising result in terms of encrypt time, and sum rate of the cryptographic system models.**

Keywords— Post quantum cryptographic, IOT networks, quantum key distribution, BBM92 Protocol and IOT networks.

I. INTRODUCTION

Amidst the prevalent landscape of data breaches and cyber threats, quantum image encryption emerges as a cutting-edge security measure. Quantum encryption techniques, particularly those rooted in quantum key distribution (QKD), guarantee a heightened level of security for the transmission and storage of images. This approach renders unauthorized access to sensitive visual data virtually impossible, proving particularly advantageous in sectors such as military intelligence and confidential corporate information exchange [1]. Quantum image processing will support satellite image processing in a reliable way which is faster make it suitable for aerospace industry [2].

Quantum mechanics supports complex calculations which are in imaginary axis values and parallelism with the base of superposition and entanglement works with the concept of qubits which are optical [4].

The existing standard technique images which are considered as classical image processing the classical images are represented in quantum states that is qubits states, individual characteristics of quantum gives way to the development of quantum image processing [5].

Image processing such as image manipulation, pixel rotation, pixel shifting and scaling which all can be performed faster with the concept of quantum image processing and in similar virtual reality and augmented reality can also be implemented with the quantum image processing [6]. Every day, the volume of data, especially the image data volume is exponentially increasing gives a way image compression and storing it for future use and same for transmission purpose it has major importance in remote sensing such as volcanic and under water images [7].

Image recognition which already plays a vital role in the field of autonomous vehicle driving needs object detection, object follow up and also facial recognition for security all the needs can be fulfilled with the mechanics of quantum, for the above requirement the quantum technologies give a faster way and with more accuracy [8].

The health care industries mainly relay on image processing for many diagnoses and prediction on certain images where decisions are taken by the doctors, certainly in medical image processing the patient health care data and all the data in-need requires security in high level which can be achieved by quantum technologies and also the speed and accuracy which can be provide by the quantum techniques are good enough to achieve the Quality of services Medical image processing which has a subset of cancer detection in various fields, quantum technology gives a promising way to do the process in a high level of accuracy[9].

A. Challenges in Quantum Image Processing

Developing quantum algorithm for security with the market edge is a high challenge, Post quantum cryptography is a subset of quantum processing, post quantum deals with security and authentication of data where more research work and brainstorming is needed. In same way balancing the technology with implementation in quantum mechanics is tough due to hardware unavailable. The noise calculation in the simulation model gives a major importance since due to noise major and minor issue can happen which impact the performance of over all system. The main challenges in quantum design are noise detection and removal and development of state of art post quantum algorithms.

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II. PROPOSED SYSTEM

In research proposing a novel technique post quantum cryptographic method which can endure the quantum attacks in nearby future, in specific proposing wavelength multiplexing embedded post quantum technique which encrypt multiple high resolution millimeter wave images, each single image resembles a pixel value of Horizontal 4160 and vertical 3120.

The wavelength multiplexing method is evaluated against the BBM92 protocol concerning processing speed, confidentiality, integrity, and non-repudiation. Post-quantum cryptographic methods operate in quantum computers, characterized by high speed, reliability, and efficient processing in qubits. These features ensure authentication and verify the identity of a system in IoT networks. The suggested method exhibits promising results in cryptographic systems. Figure 1 illustrates the fundamental structure of the future system of post-quantum cryptography and figure 2 shows the block diagram of the proposed model.

A. Quantum Cryptography

Key generation in this system user 1 chooses a random bit string, X (the raw key). User 1 also prepares a random set of quantum bits, Q (the quantum states), corresponding to each bit in X. User or system encodes each bit of a onto the quantum states using one of two possible bases: the rectilinear basis (usually denoted as Z or 0/1) or the diagonal basis (usually denoted as X or Plus or minus). The quantum states prepared by user or system can be represented as:

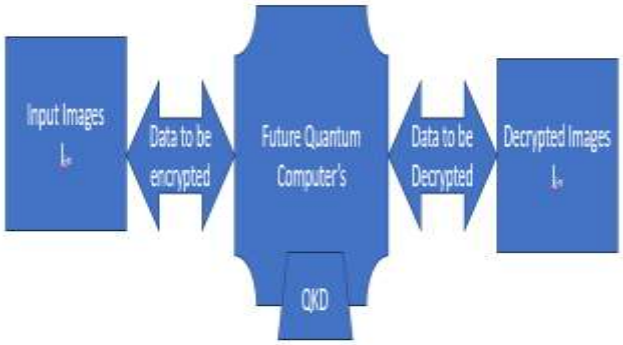


Figure 1: Basic Post Quantum Cryptography

Quantum State=$\sum_{i=1}$ alpha$_i$/b$_i$ with an upper limit N.

where N is the prepared states and b_i is the basis state and alpha are amplitude.
Quantum State measurement =$\sum_{i=1}$ $\beta_i|b_i$ with an upper limit of N.
where β_i is the measurement outcome.

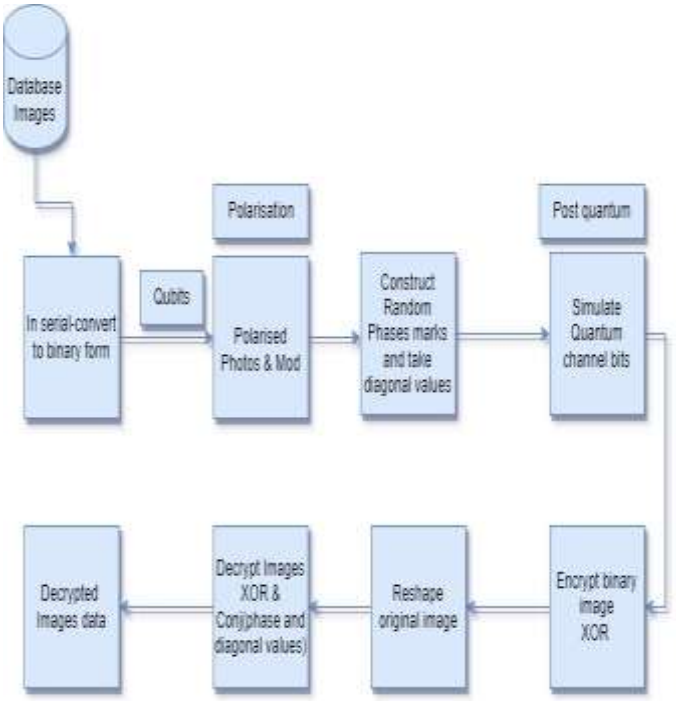


Figure 2: Proposed block diagram

B. BBM92 via Wavelength division multiplxing

Consider bipartite state in quantum, two states here BBM92 protocol employes two user assume user A and user B. User A send an information to user B using quantum networks. user B selects some qubits which is diagonal, the qubits are photos, the decoding part will take time when number of users increase in a conversational networks, same consider with wavelength division multiplexing here passive silicon photonic devices will be deployed, default this supports long distance data transmission with increase speed, wavelength division crypto method employes different wave length for each user in the channel compared to BBM92 makes a difference in the transmission medium, here each user has a single light that is photon data with two status during transmission with redundancy make the third party not able to decode the original data. Different wavelength will be used for each user, at the same time when multiple users will be multiplexed and transmission occurs makes the system more robust, a greater number of users can transmit the data at the same time with different wavelength make the system more scalability.

C.Experimental Analysis

Capturing a high-resolution image with a pixel value of 4160 pixels horizontally and 3120 pixels vertically, as indicated in a 529KB file available from reference [11]. Multiple images with similar properties are taken for the encryption process, with a single image depicted in Figure 3 and several high-resolution images shown in Figure 4. In this process, MATLAB is utilized with a support package for quantum computing. This package enables the building of circuits to implement quantum algorithms using a variety of built-in and customizable composite gates. Additionally, it facilitates the execution of quantum algorithms by connecting to quantum computing hardware through cloud services. Moreover, it allows the creation and solving of quadratic unconstrained binary optimization (QUBO) problems on a local computer, as referenced in [12].

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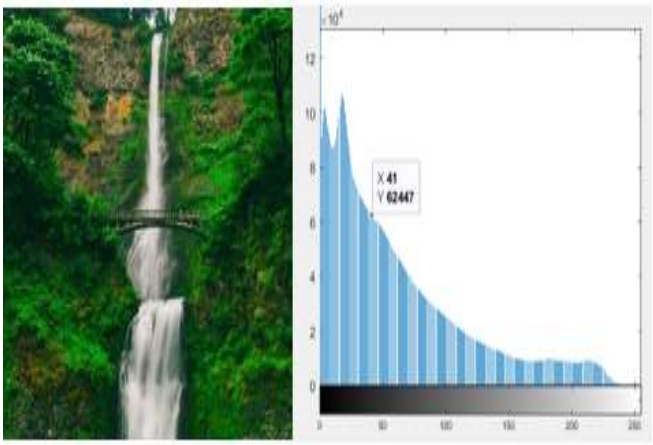


Figure 3: High Resolution Millimeter-Wave image H 4160 X V3120 Pixel Value. And its histogram value.



Figure 4: High Resolution Millimeter wave multiple images with different Pixel Value.

III. ALGORITHM

Taking the high-resolution image and converting into qubits and perform the below operation [13]. Make in note in practical quantum computers are not still available in technology, available simulation models are used. Assuming the wavelength values (ʎ) in mm, $ʎ_1 = 532e^{-6}$ and $ʎ_2 = 132e^{-6}$ assumption values can be changed as per practical scenario.

Step 1: Input images, convert the images into binary representation (I_Img).

Step 2: Define the number of Qubits (Qbits).

Step 3: Assign user 1 number of qubits and user 2 number of qubits, user 2 in measurement bases.

Step 4: user 1 encodes qubits using polarized photos,
User1_Polarisation
=Mod|(User1_Img+User2_Measurement Bases),2|

Step 5: And construct random phase masks
Random phase 1=$e^{2j(pi)*Rand(Ima_pixelvalue)}$
Random phase 2= $e^{2j(pi)*Rand(Ima_pixelvalue)}$ and taking diagonal value and encrypt.

Step 6: Simulate the Quantum channel, - Matlab support package.

Quantum_Channel_bits=User1_Polarisation+ Phase values.

Step 7: User2 data=Mod|(quantum Channel bits + user2 bases value),2|

Step 8: Encrypt_Binary_Img=XOR(binary Image+ user2 bits)
Step 9: convert the encrypt_binary_img to dec pixel value.

Step 10: Reshape to the original image.
Step 11: Decryption= conj(Encrypt_Image, ʎ Value,diagonal)

Step 12: Final_decrypt_image=user2bits+:
Encrypt_Binary_Img.

Here the polarization qubits follow wavelength division multiplying, user transmitting data between both sides with an orthogonal, and the communicating groups are in conjugate, polarization shift can cause some times loses which propagating in a wired medium, in simulation process considering the transmission medium as wireless propagating to overcomes issues such as bit error rate, when we consider the polarization in wireless medium losses are less. The polarization scheme formula below [14]

$$R_{polarization} = 1/2\left\{Q_{1,1;M^0_1}\left[1 - H\left(e_{1,1;M^0_2}\right)\right] - Q_{M^0_1} f\left(E_{M^0_1}\right) H\left(E_{M^0_1}\right)\right\} \quad (1)$$

Here $Q_{1,1}$ represent the gain of the photos and qubit error rate of the polarization system of the communicating users in equation (1). E is the inefficient function of the correlation process.

IV. OUTPUT

In default all electronic components designed are subjected to noise, commercial computers are affects via thermal noise, additive gaussian noise, noise during storage and transmission in similar quantum computers also affected by noise such as imperfections in quantum gates, thermal fluctuations, electromagnetic interference this noise can alter the quality of image being stored and transmitted. To overcome quantum noise we take two cases, one is to consider redundancy to product it from dephasing noise this make the image data better. Second is make use of quantum approximate optimization algorithm this make the devices to give more accurate values. Important note currently quantum computers are building in reality in minus temperature degree.

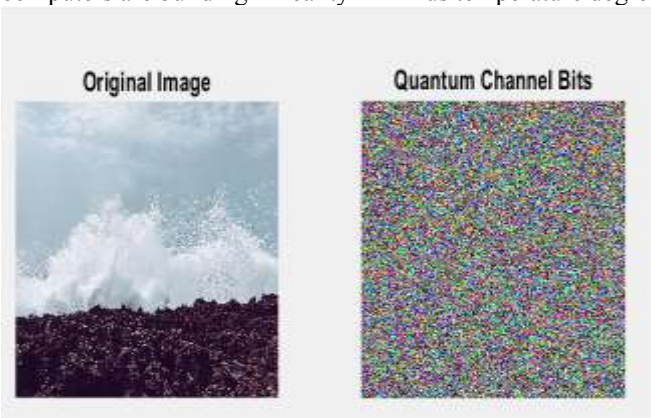


Figure 5: Original image and converted Quantum Channel bits image.

In Figure 5, the original input image is illustrated, representing the data transferred in IOT networks or other pertinent communication models. The image on the left

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showcases the distribution of quantum channel bits, wherein the input image undergoes a conversion to quantum channel bits, ensuring its security.



Figure 6: Encrypted image and decrypted image for single high-resolution image.

In Figure 6, both the encrypted and decrypted images are illustrated. The image subjected to encryption exhibits a high level of security and robustness, considering various parameters, including decryption time, key length employed in the model, and security aspects concerning external systems. It is important to note that the quantum model depicted is a simulation, as any external model currently needs to undergo simulation in a similar manner to integrate into the network.

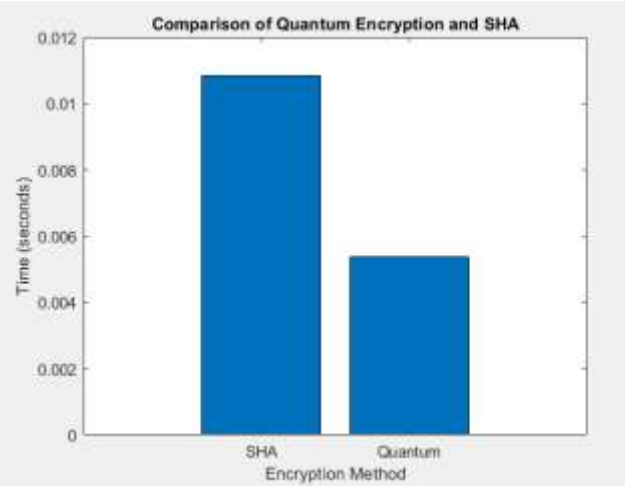


Figure 7: Execution time for image data file for Quantum and SHA method.

Figure 7 illustrates the simplified quantum simulation model. It is important to note that quantum computers do not currently exist; rather, they are part of an assumed model. In the graph above, the computing speed performance of quantum encryption with SHA is compared. The results show that the quantum method requires less time compared to existing methods.

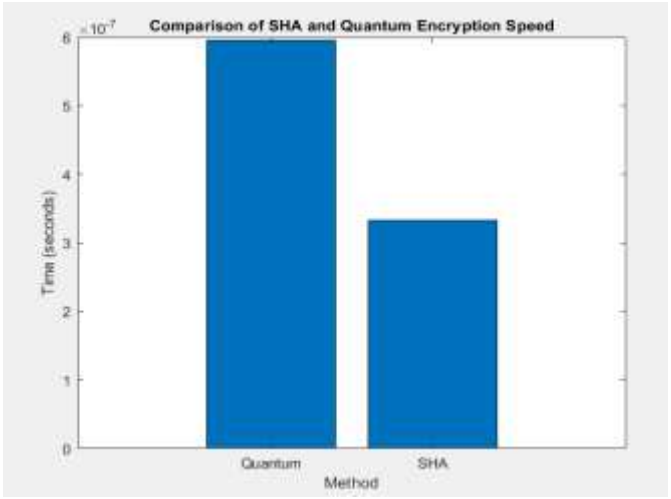


Figure 8: Basic quantum simulation model vs SHA hash computation speed.

In Figure 8, a comparison of the simulation models for the fundamental quantum process is presented. This comparison specifically evaluates the execution speed of post-quantum encryption using the SHA hash secure key model. The suggested post-quantum method demonstrates a notably high execution speed. For this simulation, key parameters include a key size of 128 bits and 100 trials for averaging, with the same 128 bits used for SHA hash.

Consider the simulation environment constructs now, will help in future quantum computers in numerous ways, firstly considering the cyber security department, same simulated mechanism can be deployed in azure services, amazon web services for quantum computing and IBM cloud services. Presently many data collections are happening in the internet web services, Internet of things, where these data are doing to be decrypted to understand the privacy of many entities. The simulated model can be a reference to implement in an entity to ensure the data privacy, data security and overcome the breaches.

V. CONCLUSION

Present era in IOT is always a threat to users for security purposes, where data security and integrity is important ans going forward when Quantum computer exists security is the essential for data when entering into the network. Post-quantum cryptographic methods, designed for the forthcoming generation of computers, exhibit significantly higher speeds compared to current computers and IoT devices. The encryption approach offered by post-quantum cryptographic methods surpasses existing methods like SHA, AES, and MD5 in terms of encryption speed, integrity, and robustness. The simulation model, implemented in the MATLAB environment using the quantum support package, ensures superior data security for high-resolution images when compared to conventional methods. The major scope of the work is to make the IOT networks future support when quantum computing entering into the era.

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